FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST – II
(Main)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C**	31.	**B**	61.	**A**
2.	**B**	32.	**C**	62.	**A**
3.	**D**	33.	**B**	63.	**C**
4.	**B**	34.	**B**	64.	**A**
5.	**A**	35.	**B**	65.	**B**
6.	**B**	36.	**D**	66.	**C**
7.	**C**	37.	**D**	67.	**D**
8.	**B**	38.	**C**	68.	**C**
9.	**C**	39.	**C**	69.	**D**
10.	**B**	40.	**A**	70.	**B**
11.	**B**	41.	**A**	71.	**B**
12.	**A**	42.	**D**	72.	**B**
13.	**B**	43.	**A**	73.	**C**
14.	**A**	44.	**B**	74.	**B**
15.	**A**	45.	**C**	75.	**A**
16.	**C**	46.	**A**	76.	**C**
17.	**D**	47.	**B**	77.	**A**
18.	**A**	48.	**D**	78.	**B**
19.	**C**	49.	**C**	79.	**C**
20.	**D**	50.	**D**	80.	**A**
21.	**B**	51.	**C**	81.	**C**
22.	**C**	52.	**A**	82.	**B**
23.	**B**	53.	**C**	83.	**D**
24.	**C**	54.	**B**	84.	**B**
25.	**A**	55.	**C**	85.	**D**
26.	**B**	56.	**D**	86.	**B**
27.	**B**	57.	**C**	87.	**A**
28.	**C**	58.	**C**	88.	**D**
29.	**D**	59.	**B**	89.	**B**
30.	**D**	60.	**C**	90.	**C**

Physics

PART – I

1. $\Delta Q_{AB} = nC_P\Delta T = 2\times\frac{5R}{2}(2T_0 - T_0) = 5RT_0$

In the process BC, PT^{-2} = constant

$PP^{-2}V^{-2}$ = constant

PV^2 = constant

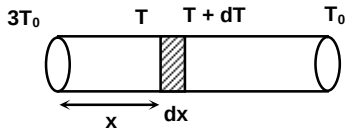
$\therefore$ molar heat capacity, $C = C_v + \frac{R}{1-x} = \frac{3R}{2} + \frac{R}{1-2}$

$C = \frac{R}{2}$

$\therefore\ \Delta Q_{BC} = nC\Delta T = 2\times\frac{R}{2}(T_0 - 2T_0) = -RT_0$

$\therefore\ \frac{|\Delta Q_{AB}|}{|\Delta Q_{BC}|} = \frac{5RT_0}{RT_0} = 5$

2. In steady state



$\frac{\Delta Q}{\Delta t} = -kA\frac{dT}{dx}$

$\frac{\Delta Q}{\Delta t} = -\alpha TA\frac{dT}{dx}$

$\frac{\Delta Q}{\Delta t}\int_0^{\ell} dx = -\alpha A\int_{3T_0}^{T_0} TdT$

$\frac{\Delta Q}{\Delta t}\ell = 4\alpha AT_0^2$ …(i)

Similarly,

$\frac{\Delta Q}{\Delta t}\int_0^{\ell/2} dx = -\alpha A\int_{3T_0}^{T} TdT$

$\frac{\Delta Q}{\Delta t}\frac{\ell}{2} = \alpha A\frac{(9T_0^2 - T^2)}{2}$ …(ii)

Dividing (ii) by (i)

$9T_0^2 - T^2 = 4T_0^2$

$T^2 = 5T_0^2$

$T = \sqrt{5}T_0$

3. Given PT^3 = constant

$\frac{T}{V}T^3 = \text{constant}$

$\therefore\ V = kT^4$

The coefficient of thermal expansion is

$\gamma = \frac{dV}{VdT} = \frac{4kT^3}{kT^4} = \frac{4}{T}$

4. Equilibrium temperature of mixture is 0°C.
Total heat released by water = $m_w S_w \Delta\theta = 5 \times 1 \times 40 = 200$ kcal
Heat taken by ice to increase its temperature from $-20°C$ to 0°C = $4 \times 0.5 \times 20 = 40$ kcal
$\therefore$ mass of ice melted = $\frac{200-40}{80} = \frac{160}{80} = 2$ kg
$\therefore$ water content in the equilibrium mixture = 5 + 2 = 7 kg

5. Given
$C = C_V + \alpha T^2$
$$C_V + \frac{RT}{V}\frac{dV}{dT} = C_V + \alpha T^2$$
$$\int \frac{\alpha T}{R} dT = \int \frac{dV}{V} + \ell nk$$
$$\frac{\alpha T^2}{2R} = \ell n(kV)$$
$$kV = e^{\frac{\alpha T^2}{2R}}$$
$$\therefore\ Ve^{-\left(\frac{\alpha T^2}{2R}\right)} = \text{constant}$$

6. The net electric potential due to external charge inside the cavity and the induced charge on the surface of the cavity at any point outside the cavity is zero. If 'V' is the electric potential due to induced charge on the surface of cavity at point 'P'.
$$V + \frac{kq}{5a} = 0$$
$$\therefore\ V = -\frac{kq}{5a}$$

7. $$-mS\frac{dT}{dt} = \sigma A T^4$$
$$-\rho\frac{4}{3}\pi R^3 S\frac{dT}{dt} = \sigma 4\pi R^2 T^4$$
$$-\int_{T_0}^{T_0/2}\frac{dT}{T^4} = \frac{3\sigma}{\rho SR}\int_0^t dt$$
$$\left[\frac{1}{3T^3}\right]_{T_0}^{T_0/2} = \frac{3\sigma}{\rho SR}t$$
$$\left(\frac{8}{T_0^3}\right) - \frac{1}{T_0^3} = \left(\frac{9\sigma t}{\rho SR}\right)$$
$$\therefore\ t = \frac{7\rho SR}{9\sigma T_0^3}$$

8. The electric field due to $\vec{P}_1$ at the position of $\vec{P}_2$ is
$$E_1 = \frac{KP_1}{r^3}$$
$\therefore$ force of inter action between the dipoles is

$$F = P_2\left|\frac{dE_1}{dr}\right|$$

$$F = \frac{3kP_1P_2}{r^4}$$

9. $$dS = \frac{dQ}{T}$$

$$\int dS = nc_P\int_{T_0}^{2T_0}\frac{dT}{T}$$

$$\Delta S = 4\times\frac{5R}{2}\ell n2 = 10R\ell n2 = 10\times\frac{25}{3}\times 0.69$$

$$25\times 2.3 = 57.5\ J/k$$

10. The electric flux through the right end plane surface.

$$\phi_{PS} = \frac{q}{4\pi\varepsilon_0}\times 2\pi(1-\cos\theta) = \frac{q}{2\varepsilon_0}\left(1-\frac{4}{5}\right)$$

$$\phi_{PS} = \frac{q}{10\varepsilon_0}$$

Hence, the electric flux through the curved surface of the cylinder is

$$\phi_{CS} = \frac{q}{2\varepsilon_0} - \phi_{PS}$$

$$= \frac{q}{2\varepsilon_0} - \frac{q}{10\varepsilon_0} = \frac{4q}{10\varepsilon_0}$$

$$\phi_{CS} = \frac{2q}{5\varepsilon_0}$$

11. $$U_i = \frac{k(2Q)^2}{2a} + \frac{kQ^2}{4a} + \frac{k(2Q)Q}{2a}$$

$$= \frac{2kQ^2}{a} + \frac{5kQ^2}{4a}$$

$$U_i = \frac{13kQ^2}{4a}$$

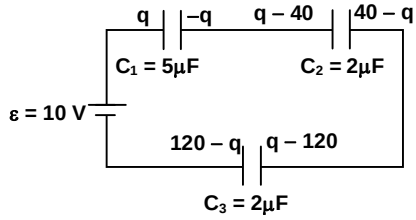
$$U_f = \frac{k(3Q)^2}{4a} = \frac{9kQ^2}{4a}$$

$\therefore$ Total heat dissipated, $\Delta H = U_i - U_f$

$$= \frac{13kQ^2}{4a} - \frac{9kQ^2}{4a}$$

$$= \frac{4kQ^2}{4a} = \frac{kQ^2}{a}$$

12. $\frac{q}{5}+\left(\frac{q-40}{2}\right)+\left(\frac{q-120}{2}\right)=10$



$\frac{6q}{5}=90$

q = 75 μC

∴ magnitude of charge flown through the battery is

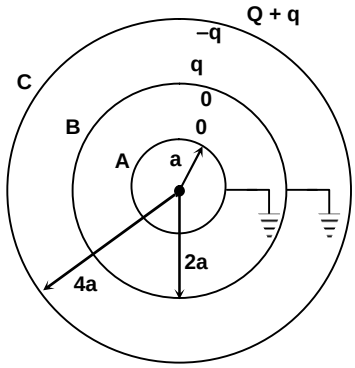
$\Delta q = 100 - 75 = 25\ \mu C$

13. The electric potential of shell B is zero

$V_B = 0$

$\frac{kq}{2a}+\frac{kQ}{4a}=0$

$\therefore\ q=-\frac{Q}{2}$



14. The cell will deliver maximum power to the network if $R_{eq} = r$

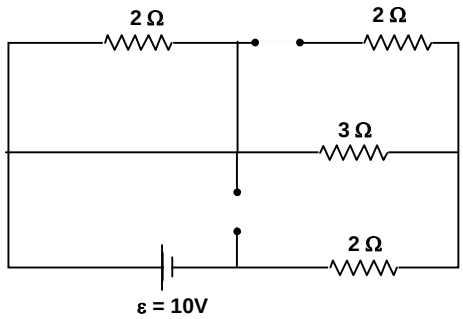
$\Rightarrow \frac{3R}{2}=r$

$\therefore\ R=\frac{2r}{3}=\frac{2}{3}\Omega$

15. The equivalent circuit just after closing the switch 'S' is

$R_{eq} = 3 + 2 = 5\ \Omega$

$\therefore\ I_0=\frac{\varepsilon}{R_{eq}}=\frac{10}{5}=2A$



16. Since the particle exists the magnetic field region with a velocity $\vec{v}=(3\hat{i}+3\sqrt{3}\hat{j})$ m/s

$\tan\theta=\frac{3\sqrt{3}}{3}$

$\therefore\ \theta=\frac{\pi}{3}$

The angular displacement of the particle in the magnetic field region is $\theta=\frac{\pi}{3}$

Angular velocity, $\omega=\frac{qB}{m}$

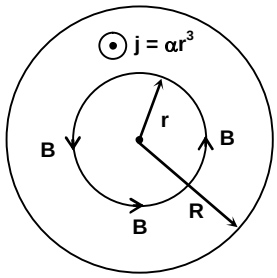
$\therefore\ t=\frac{\theta}{\omega}=\frac{\pi m}{3qB}$

17. Since $\vec{B}$ is constant in the region
The net magnetic force experienced by the conducting wire MN is
$\vec{F} = I\left(\vec{L}_{eff} \times \vec{B}\right)$
Where, $\vec{L}_{eff} = 6R\hat{i}$
$\vec{F} = I\left[6R\hat{i} \times B(-\hat{k})\right]$
$F = 6BIR(\hat{j})$

18. The net magnetic moment of the loop is
$\vec{M} = I\ell^2(\hat{i} + \hat{k})$
The net magnetic torque experienced by the loop is
$\vec{\tau} = \vec{M} \times \vec{B}$
$= I\ell^2(\hat{i} + \hat{k}) \times B\hat{j}$
$\vec{\tau} = BI\ell^2(\hat{k} - \hat{i})$

19. The net magnetic field at the centre 'O' of the circular arc is
$B_0 = \frac{\mu_0 I}{4\pi R} + \frac{3\mu_0 I}{8R} - \frac{\mu_0 I}{8\pi R}$
$= \frac{\mu_0 I}{8\pi R} + \frac{3\mu_0 I}{8R}$
$B_0 = = \frac{\mu_0 I}{8R}\left(\frac{1+3\pi}{\pi}\right)$

20. When $r \le R$



$B2\pi r = \mu_0 \int_0^r j2\pi r dr$
$Br = \frac{\mu_0 \alpha r^5}{5}$
$B = \frac{\mu_0 \alpha r^4}{5}$, $r \le R$
when $r > R$
$B2\pi r = \mu_0 \int_0^R j2\pi r dr$
$Br = \mu_0 \int_0^R \alpha r^3 r dr$
$Br = \frac{\mu_0 \alpha R^5}{5}$
$\therefore\ B = \frac{\mu_0 \alpha R^5}{5r}$, $r > R$
at $r = \frac{R}{2}$, $B_1 = \frac{\mu_0 \alpha R^4}{80}$
at $r = 2R$, $B_2 = \frac{\mu_0 \alpha R^4}{10}$
$\therefore\ \frac{B_2}{B_1} = 8$

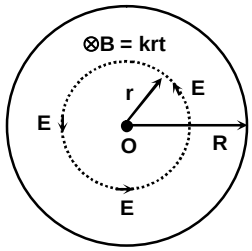
21. The emf induced between the points 'A' and 'D' is

$\varepsilon = \vec{B}\cdot(\vec{\ell}_{eff}\times\vec{v})$

$= Ba\sqrt{2}v\sin 45°$

$= Ba\sqrt{2}\dfrac{v}{\sqrt{2}} = Bva = 2B\omega a^2$

22. When $r \le R$



$E2\pi r = \dfrac{d}{dt}\int_0^r B2\pi r dr$

$Er = \dfrac{d}{dt}\left(\dfrac{ktr^3}{3}\right)$

$E = \dfrac{kr^2}{3}$, when $r \le R$

When $r > R$

$E2\pi r = \dfrac{d}{dt}\left(\int_0^R B2\pi r dr\right)$

$Er = \dfrac{d}{dt}\left(\int_0^R krtrdr\right)$

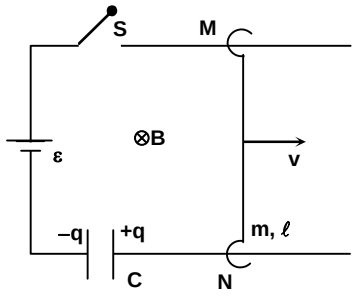
$E = \dfrac{kR^3}{3r}$, where $r > R$

$\therefore$ at $r = \dfrac{R}{2}$, $E_1 = \dfrac{kR^2}{12}$

at $r = 2R$, $E_2 = \dfrac{kR^2}{6}$

$\therefore \dfrac{E_2}{E_1} = 2$

23. $\dfrac{q}{C} + Bv\ell = \varepsilon$ …(i)



$q = C(\varepsilon - Bv\ell)$

Now, $m\dfrac{dv}{dt} = Bi\ell$

$mdv = B\ell idt$

$m\int_0^v dv = B\ell\int_0^q dq$

$mv = B\ell q$

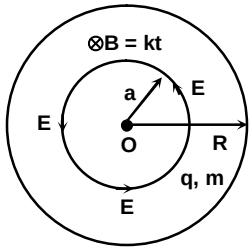
$\therefore q = \dfrac{mv}{B\ell}$ …(ii)

From (i) and (ii)

$\dfrac{mv}{B\ell} = C\varepsilon - CB\ell v$

$$\Rightarrow v = \frac{CB\ell\varepsilon}{(m + CB^2\ell^2)}$$

24. $E2\pi r = \frac{d}{dt}(kt\pi r^2)$



$$E = \frac{kr}{2},\ r \le R$$

at r = a

$$E = \frac{ka}{2}$$

To keep the ring stationary

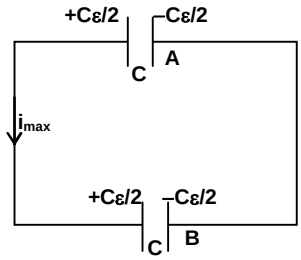
$qEa = \mu mga$

$$\frac{qka}{2} = \mu mg$$

$$\therefore\ \mu = \frac{kqa}{2mg}$$

$$\mu_{min} = \frac{kqa}{2mg}$$

25. When the current through the inductor is maximum



$$V_L = L\frac{di}{dt} = 0$$

Hence capacitors A and B will be in parallel

Now, using conservation of energy

$$\frac{1}{2}C\varepsilon^2 = 2\times\frac{1}{2}C\frac{\varepsilon^2}{4} + \frac{1}{2}Li_{max}^2$$

$$\frac{C\varepsilon^2}{4} = \frac{1}{2}Li_{max}^2$$

$$i_{max} = \varepsilon\sqrt{\frac{C}{2L}}$$

26. $L_1\frac{di_1}{dt} = L_2\frac{di_2}{dt}$

$$L_1\int_0^{i_1} di_1 = L_2\int_0^{i_2} di_2$$

$L_1i_1 = L_2i_2$ …(i)

$i_1 + i_2 = i$ …(ii)

$$i_1\left(1 + \frac{L_1}{L_2}\right) = i$$

$$\Rightarrow i_1 = \left(\frac{L_2}{L_1 + L_2}\right)i \text{ and } i_2 = \left(\frac{L_1}{L_1 + L_2}\right)i$$

$$\therefore\ \frac{U_1}{U_2} = \frac{\frac{1}{2}L_1i_1^2}{\frac{1}{2}L_2i_2^2} = \frac{L_1L_2^2}{L_2L_1^2} = \frac{L_2}{L_1}$$

27. Rms current through resistor 'R_1' is

$$I_{1(rms)} = \frac{50}{\sqrt{2}\times 10\sqrt{2}} = \frac{5}{2}A$$

$\therefore$ Average power developed in the resistor 'R_1' is

$$P_1 = I_{1(rms)}^2 R_1 = \left(\frac{5}{2}\right)^2 10 = \frac{250}{4} = 62.50\,W$$

28. Current through the potentiometer wire is

$$I = \frac{6}{R+5}$$

$$\therefore\ V = \frac{6\times 3}{(R+5)}$$

$$2 = \frac{18}{R+5}$$

$R + 5 = 9$

$R = 4\ \Omega$

29. Let the initial charge on the spherical conductor 'A' is Q.

$$\therefore\ \frac{kQ}{a} = V$$

$\Rightarrow$ $kQ = Va$ …(i)

When switch 'S' is closed, the total charge 'Q' will appear on the outer surface of shell B.

$\therefore$ Final potential of the spherical conductor 'A' is

$$V' = \frac{kQ}{6a} = \frac{Va}{6a} = \frac{V}{6}$$

30. Given process is

VT^2 = constant

VP^2V^2 = constant

$PV^{3/2}$ = constant

$\therefore$ molar heat capacity

$$C = C_V + \frac{R}{1-x} = \frac{5R}{2} + \frac{R}{1-\frac{3}{2}}$$

$\Rightarrow C = R/2$

$\therefore$ Heat supplied to the gas is

$$\Delta Q = nC\Delta T = 2\times\frac{R}{2}\times 300 = 300\,R$$

Increase in internal energy of the gas is

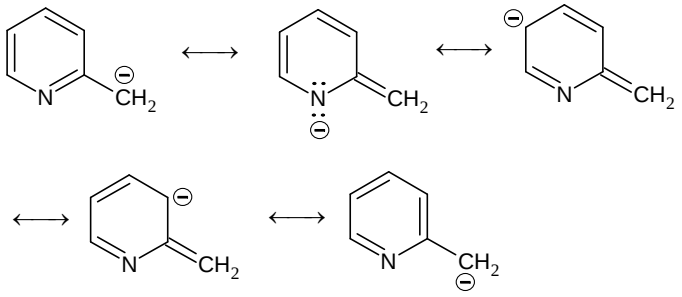
$$\therefore\ \Delta U = nC_V\Delta T = 2\times\frac{5R}{2}\times 300 = 1500R$$

$$\therefore\ \text{The ratio } \frac{\Delta U}{\Delta Q} = \frac{1500R}{300R} = 5$$

Chemistry

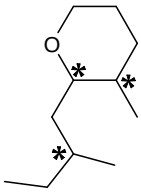
PART – II

31.



32. The carboxylic acid group stabilizes the conjugate base and – I of other groups at closest distance.

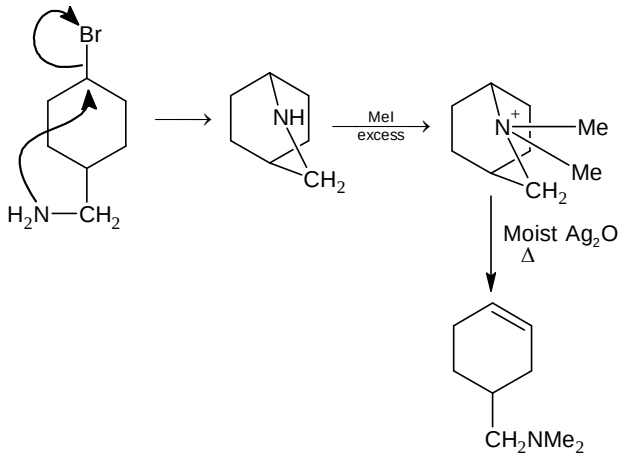
34. Total chiral centres = 3
$2^n = 2^3 = 8$



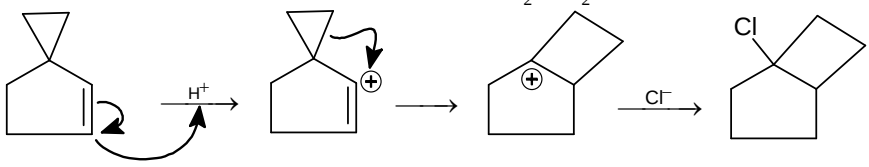
35. In B all alkyl groups are at equatorial positions so most stable.
Most stable will have least heat of combustion.

37. Glyptal is made from polymerization of ethylene glycol and pthalic acid.

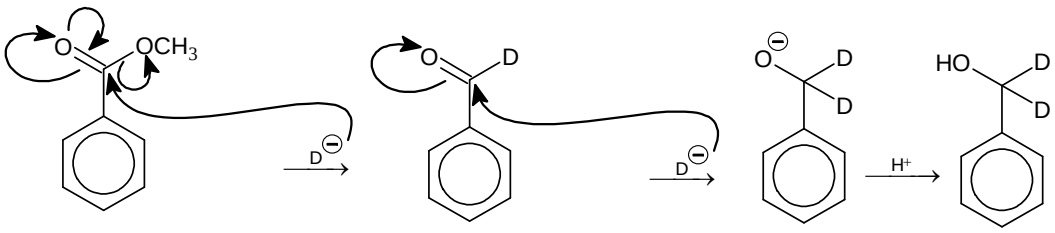
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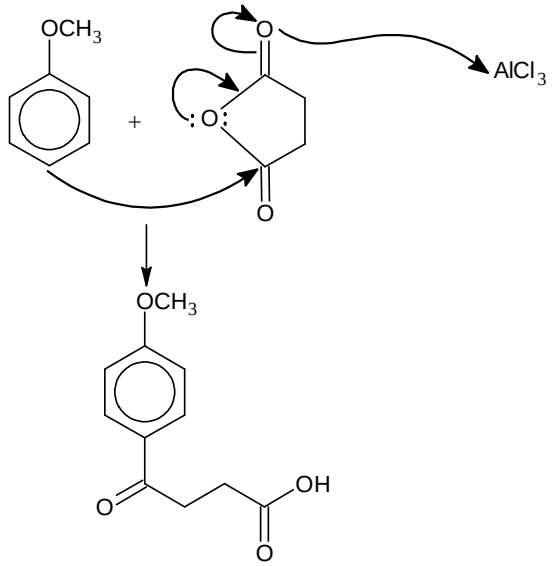
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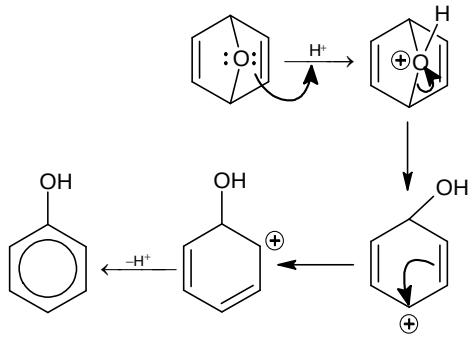
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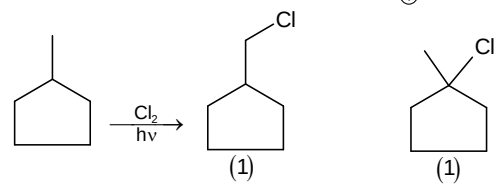
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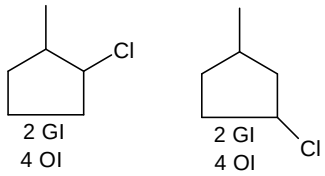
46.

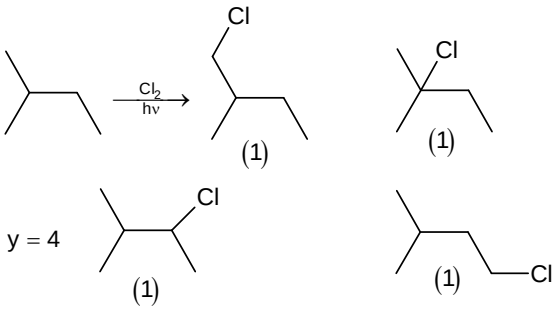


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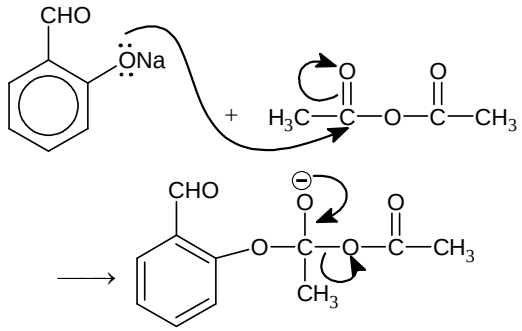


x = 10

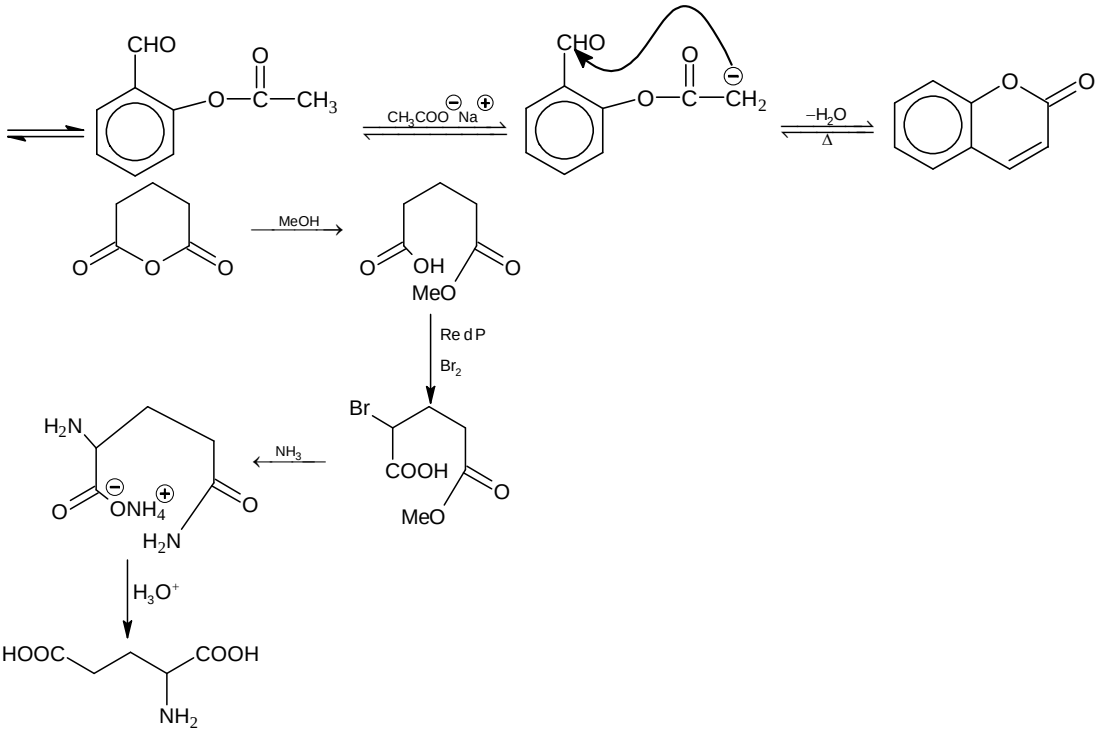



Cl
Cl_2
hν
(1)
Cl
(1)
y = 4
Cl
(1)
(1)
Cl

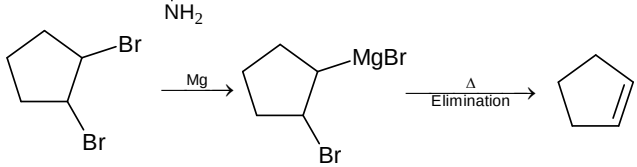
48.


CHO
ONa
+
$H_3C-C-O-C-CH_3$
CHO
CH_3

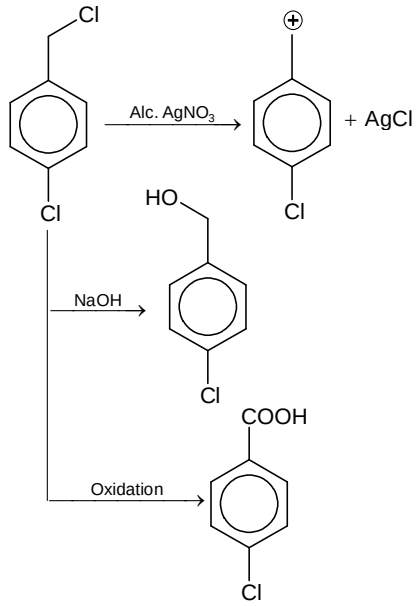
50.


CHO
$CH_3COO^{\ominus}\ Na^{\oplus}$
CHO
$-H_2O$
Δ
MeOH
OH
MeO
Re d P
Br_2
Br
COOH
MeO
NH_3
H_2N
ONH_4
H_2N
H_3O^+
HOOC
COOH
NH_2

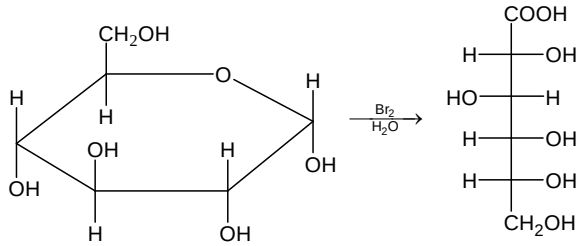
51.


Br
Br
Mg
MgBr
Br
Δ
Elimination

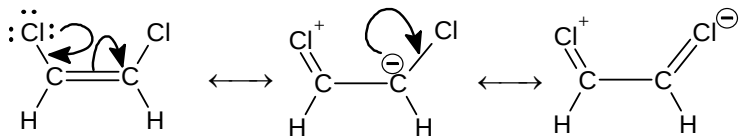
52.



53.

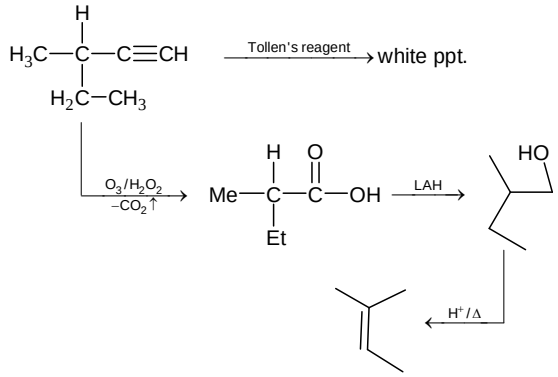


55.



In cis form attraction between both chlorine leads to stability.

56.



57. (ii), (vi) will not give Cannizzaro test.

58. Rate of reaction depends upon migration aptitude.

60. Intermolecular aldol reaction.

Mathematics

PART – III

61. $\tan(\alpha+\beta) = -\frac{p}{1-q} = \frac{p}{q-1}$

$\frac{1}{1+\tan^2(\alpha+\beta)}\left(\tan^2(\alpha+\beta) + p\tan(\alpha+\beta) + q\right)$

$= \frac{1}{1+\frac{p^2}{(q-1)^2}}\left(\frac{p^2}{(q-1)^2} + \frac{p^2}{q-1} + q\right)$

$= \frac{1}{p^2+(q-1)^2}\left(p^2 + p^2(q-1) + q(q-1)^2\right) = q$

62. $f(x) = (x-1)(x-\cos\theta)(x-\sin\theta)$
$\alpha^2 + \beta^2 + \gamma^2 = 1 + \sin^2\theta + \cos^2\theta = 2$

63. $x = \sqrt{2} - 1 \Rightarrow (x+1)^2 = 2$
$\Rightarrow x^4 + 3x^3 - 3x^2 - 9x + 6 = (x^2 + x - 4)(x^2 + 2x - 1) + 2 = 2$

64. $x = r\cos\theta,\ y = r\sin\theta$
$\Rightarrow r^2 \sin\theta\cos\theta\, r^2(\cos^2\theta - \sin^2\theta) = r^2$
$\Rightarrow r^2 = \frac{4}{\sin 4\theta} = 4\ \text{cosec}\ 4\theta \geq 4$

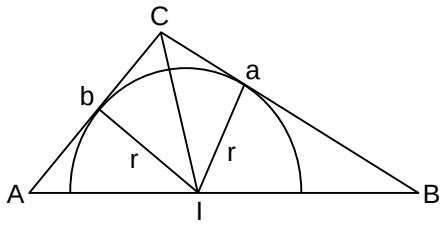
65. $\sec^2\frac{\pi}{8} + \text{cosec}^2\frac{\pi}{8} = \sec^2\frac{\pi}{8}\text{cosec}^2\frac{\pi}{8}$
$\Rightarrow -a = b \Rightarrow a + b = 0$

66. $x = \frac{2ab}{a+b}\cos\frac{C}{2}$

$\frac{1}{2}ra + \frac{1}{2}rb = \frac{1}{2}ab\sin C$

$\Rightarrow r = \frac{2\cdot\frac{1}{2}ab\sin C}{a+b} = \frac{2ab\sin\frac{C}{2}\cos\frac{C}{2}}{a+b} = x\sin\frac{C}{2}$

$r = \frac{2\Delta}{a+b} \Rightarrow r = \frac{abc}{4R^2(\sin A + \sin B)}$



67. $\cos x = 0$ or $\sin\left(x + \frac{1}{x}\right) = 0$

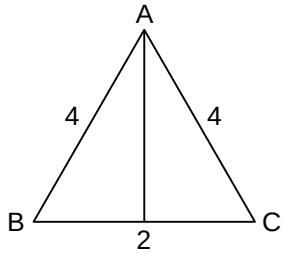
$\Rightarrow x = \frac{\pi}{2}$ or $x + \frac{1}{x} = n\pi \Rightarrow$ infinite solution

68. $f(x) = \frac{\left(\text{cosec}^2 x - 1\right)^2}{\left(\text{cosec}^2 x - 1\right) + 1 - \cot x + \cot x} = \frac{\cot^4 x}{1+\cot^2 x} = 0\ ;\ x \in R - n\pi,\ n \in I$

$\Rightarrow x = (2n-1)\frac{\pi}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, ..., \frac{199\pi}{2}$

Sum = $\frac{\pi}{2}(1+3+5+.....+199) = 5000\pi$

69. $p = 1, a_2 = 4 \Rightarrow b = 4$
$p = 2, a_3 = 4 \Rightarrow c = 4$
$\Rightarrow \Delta$ is isosceles
$\Delta = \sqrt{15}$
$r_1 = \frac{\Delta}{s-a} = \frac{\sqrt{15}}{3}$, $r_2 = \frac{\Delta}{s-b} = \sqrt{15} = r_3$
$\Rightarrow r_2 = r_3 = 3r_1$

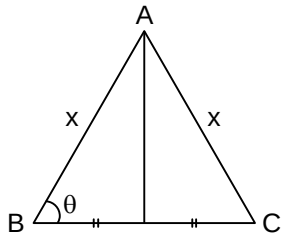


70. $\tan^{-1}\left(\frac{4}{4r^2+3}\right) = \tan^{-1}\left(\frac{1}{r^2+\frac{3}{4}}\right) = \tan^{-1}\left(\frac{1}{1+\left(r+\frac{1}{2}\right)\left(r-\frac{1}{2}\right)}\right) = \tan^{-1}\left(r+\frac{1}{2}\right) - \tan^{-1}\left(r-\frac{1}{2}\right)$

$\sum_{r=1}^{n}\tan^{-1}\left(\frac{4}{4r^2+3}\right) = \tan^{-1}\left(n+\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{2}\right)$

$\Rightarrow \sum_{r=1}^{\infty}\tan^{-1}\left(\frac{4}{4r^2+3}\right) = \tan^{-1}2$

71. $A = \frac{1}{2}(2x\cos\theta)x\sin\theta = \frac{1}{2}x^2\sin 2\theta$
$Max(A) = \frac{x^2}{2}$



72. Lines intersect at $\left(-3, \frac{3}{2}\right)$ which is centre of required circle. Given circle $x^2 + y^2 - 4x - 3y = 0$ whose centre is $\left(2, \frac{3}{2}\right)$, $r = \frac{5}{2}$
For just sufficient circle $d = |r_1 - r_2|$
$\Rightarrow \left|\frac{5}{2} - r\right| = 5 \Rightarrow r = \frac{15}{2} \Rightarrow 9 + \frac{9}{4} + k = \frac{225}{4} \Rightarrow k = 45$

73. If acute angle = $\theta \Rightarrow \cos(\pi - \theta) = \frac{a^2 + a^2 - d_1^2}{2a^2}$
$\Rightarrow 2a^2 - d_1^2 = -2a^2\cos\theta \Rightarrow d_1 = \sqrt{2a^2 + 2a^2\cos\theta} = 2a\cos\frac{\theta}{2}$
Similarly, $d_2 = \sqrt{2a^2(1-\cos\theta)} = 2a\sin\frac{\theta}{2}$ and $d_1d_2 = a^2$
$\Rightarrow 4\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 1 \Rightarrow \sin\theta = \frac{1}{2}$

74. Let $\log_{\sqrt{3}} \tan x = t \Rightarrow t\sqrt{3+\frac{2}{t}} = -1,\ t < 0$

$\Rightarrow t^2\left(3+\frac{2}{t}\right) = 1 \Rightarrow 3t^2 + 2t - 1 = 0 \Rightarrow t = -1, \frac{1}{3}$

$\Rightarrow \log_{\sqrt{3}} \tan x = -1 \Rightarrow \tan x = \frac{1}{\sqrt{3}} \Rightarrow x = \frac{\pi}{6}$

75. $\tan B + \tan C = \frac{\sin(B+C)}{\cos B \cos C} = \frac{2R\sin A}{2R\cos B\cos C} = \frac{a}{2R\cos B\cos C} = \lambda$

$\Rightarrow$ 2R cos B cos C = k

$\Rightarrow$ Locus of orthocentre is a line parallel to BC

76. $2 - a = SP = \frac{1}{2} \Rightarrow a = \frac{3}{2} \Rightarrow 2a = 3$

77. (1, –1) lies on y + x = 0. Hence only one rational point

78. $\sqrt{x^2+y^2+2y+1} = \sqrt{x^2+(y+1)^2}$ = distance between (x, y) and (0, –1)

The distance from (x, y) to x – 2y + 3 = 0 is $\frac{|x-2y+3|}{\sqrt{5}}$

$\Rightarrow \sqrt{x^2+(y+1)^2} = \frac{|x-2y+3|}{\sqrt{5}} \cdot \sqrt{\frac{5}{\lambda}}$

$\Rightarrow e = \sqrt{\frac{5}{\lambda}}$ for hyperbola $\sqrt{\frac{5}{\lambda}} > 1$

$\Rightarrow \lambda < 5$

79. $a + b + \sqrt{a^2+b^2} = \frac{ab}{2}$

$\Rightarrow b = \frac{4(a-2)}{a-4}$ for a = 5, a = 6, b = 8

$\Rightarrow \Delta = \frac{1}{2}ab = 24$

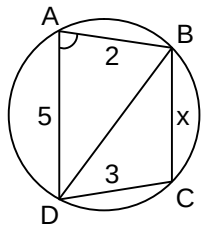
80. $BD^2 = 2^2 + 5^2 - 2\cdot2\cdot5\cdot\frac{1}{2}$

$= 3^2 + x^2 - 2\cdot3\cdot x\left(-\frac{1}{2}\right)$

$\Rightarrow 19 = x^2 + 3x + 9$

$\Rightarrow x^2 + 3x - 10 = 0$

$\Rightarrow x = 2$



81. $\Delta = \frac{1}{2}\cdot 4\sqrt{\frac{2}{5}}\cdot\frac{|2\cos\theta - \sin\theta|}{\sqrt{2}} \le 2$

82. $\frac{PA}{PB} = \frac{\sqrt{10}-2\sqrt{2}}{\sqrt{10}+2\sqrt{2}} \Rightarrow$ AB is a diameter

83. $2b\tan\theta = \sqrt{a^2\sec^2\theta + b^2\tan^2\theta}$

$\Rightarrow e^2 > 1 + \frac{1}{3}$

84. $\frac{PA + PB}{2} \geq \sqrt{PA \cdot PB}$

$\Rightarrow PA + PB \geq 2PT$

85. $a\left(x + \frac{y}{2}\right) + c\left(\frac{y}{2} + 1\right) = 0$

$\Rightarrow \alpha = 1, \beta = -2$ and $4(\alpha + \beta) = \alpha^2 + \beta^2 - \gamma - 1$

$\Rightarrow \gamma = 8$

$\alpha + \beta + \gamma = 7$

86. Equation of tangent is $\frac{x}{3\sqrt{3}}\cos\theta + y\sin\theta = 1$

$S = \frac{3\sqrt{3}}{\cos\theta} + \frac{1}{\sin\theta}$. It is minimum at $\theta = \frac{\pi}{6}$

87. Orthocentre (h, k) = $\left(2a + a\left(t_1^2 + t_2^2 + t_1t_2\right), -at_1t_2\left(t_1 + t_2\right)\right)$ where $t_2 = -\frac{1}{t_1}$

$\Rightarrow k = a(t_1 + t_2)$

$h = 2a + a\left(\left(t_1 + t_2\right)^2 - t_1t_2\right) = 2a + a\left(\frac{k^2}{a^2} + 1\right)$

$ah = 3a^2 + k^2$

88. Sides of squares will be perpendicular tangents to the ellipse. So vertices of square will lie on diameter circle

$\Rightarrow$ Diameter of circle = $2\sqrt{a^2 - 3 + a + 4} = \sqrt{2a^2 + 2a^2}$

$\Rightarrow a^2 + a + 1 = a^2$

$\Rightarrow a = -1$

But $a^2 > 3$ and $a > -4$

$\Rightarrow a \in \phi$

89. $\angle C = \frac{\pi}{2}$, A = 15º, B = 75º

90. $\frac{1}{(OA)^2} + \frac{1}{(OB)^2} = \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{4} + 1 = \frac{5}{4}$